

Five Tools for Verified Gaps in CRISPR Guide-RNA and Off-Target Modeling: Cross-Dataset Generalization, Calibrated Cleavage Probabilities, Transfer Learning for pegRNAs, PAM-Variant Awareness, and Chromatin-Context Scoring

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Abstract

Public CRISPR tooling concentrates on a narrow slice of the editing problem. We attack five evidence-backed gaps with working tools benchmarked on real public data: (1) cross-dataset collapse of guide efficacy models, which we quantify on Doench-2016 FC+RES, Doench V1, the DeepHF U2OS screen and the DeepSpCas9 HEK293T dataset (Kim 2019); (2) absence of calibrated probabilities from off-target scorers, addressed with isotonic/Platt calibration on 25,632 crisprSQL measurements; (3) pegRNA data scarcity, attacked with transfer learning from Cas9 efficacy data to the 22,956-pegRNA PRIDICT2 dataset; (4) PAM-variant-blind off-target scoring; (5) chromatin-context-blind off-target scoring. All claims are bounded by held-out, group-aware evaluation; negative results are reported. Beyond the five gap tools, the analysis surfaces a *discovery*: a target-inherent transfer residual – the per-guide error of a source-trained model replicates at $\rho = 0.86\text{--}0.90$ across three independent nuclease screens ($n = 53,937$ matched guides), more stably than the biological signal itself ($\rho \approx 0.73$). The residual is predictable from sequence and chromatin features and yields **PORTA**, a portability correction that rescues cross-dataset transfer from $\rho = 0.080$ to 0.524 (SpCas9-HF1) and 0.052 to 0.562 (eSpCas9) with zero destination labels, and exposes a per-guide portability score no current tool reports. On calibrated off-target scoring, five fully-seeded guide-grouped splits of crisprSQL yield a meta-finding: every model’s AUROC swings by ~ 0.2 across splits and inits (GNN $0.45\text{--}0.69$), dwarfing model gaps – single-split published comparisons on negative-scarce corpora are irreproducible by construction. The strongest model is a position-specific logistic regression (0.692 ± 0.154), beating the GNN (0.609 ± 0.084) on 4/5 identical splits; no deep model separates from the classical baselines (CFD 0.552, MIT 0.516). A natural per-study routing hypothesis (GNN for low-prevalence GUIDE-seq-style studies, MIT otherwise) is *falsified* under leave-one-out evaluation: the apparent per-study GNN wins rest on as few as two positives, and the robust policy on this corpus is the MIT baseline everywhere (Section 6).

All artifacts – code, 67-test offline suite, data pipelines, figures, and this paper’s numbers – regenerate from committed scripts.

1 Introduction

Computational CRISPR tooling has converged on a narrow problem shape: SpCas9 on-target efficacy in a handful of human cell lines, and off-target *ranking* scores for NGG-PAM targets. Auditing the public tool and dataset landscape surfaces five gaps, each backed by dataset-level evidence: (1) efficacy models are rarely evaluated across independent assays, and reported performance collapses when they are; (2) the widely used MIT and CFD off-target scores are uncalibrated ranking heuristics – no published tool returns a probability of cleavage with an error bound; (3) prime-editing pegRNA design models are data-scarce relative to Cas9 nuclease models by two orders of magnitude (22,956 measured pegRNAs vs. millions of Cas9 guide measurements), yet transfer from Cas9 data is essentially untried in public tools; (4) off-target scorers are trained and validated almost exclusively on NGG-PAM off-targets even though 47% of measured off-targets in crisprSQL carry non-NGG PAMs; (5) chromatin context, known to modulate cleavage genome-wide, is absent from the standard sequence-only scoring stack despite crisprSQL shipping CTCF, DNase, RRBS, H3K4me3 and DRIP tracks for every one of its 25,632 measured pairs. We build one tool per gap and report honest, group-aware held-out numbers, including where the gap resisted us.

Contributions. (i) A measurement harness for cross-dataset generalization of efficacy models, with the first cross-family transfer matrix on Doench/DeepHF (Section 5). (ii) A calibrated multi-study off-target benchmark with full split/init determinism, establishing that negative-scarce corpora cannot adjudicate between scorers on single splits – and that per-study apparent wins, including our own, are positive-count artifacts (a study-routing hypothesis is stated and falsified, Section 6). (iii) A controlled study of Cas9→pegRNA transfer, establishing an honest negative at three scales in two cell lines (Section 7). (iv) PAM- and chromatin-aware off-target channels; five-seed replication reverses our own single-split finding – no channel’s delta exceeds split noise, and PAM alone *hurts* on average (Section 8). (v) A named, quantified, falsifiable discovery – the target-inherent transfer residual – and PORTA, the correction it enables (Section 12). (vi) A reproducible codebase: 40 hermetic tests, public data only, every figure regenerated from result files (Appendix A).

Roadmap. Section 2 surveys prior tools, Section 3 the datasets, Section 4 the mathematical objects (12 numbered equations, 4 proved propositions), Sections 5–8 the five gaps, Sections 9–10 methods and protocols, Section 11 results with bootstrap support, Section 12 the discovery and PORTA, Section 13 discussion, Sections 14–15 limitations and tool documentation.

2 Related work

2.1 gRNA efficacy models

On-target activity prediction began with position-specific sequence models. Rule Set 1 (Doench et al. 2014) introduced position-dependent dinucleotide features in a logistic model; Rule Set 2 / Azimuth (Doench et al. 2016) added position-independent dinucleotides, GC counts, and thermodynamic accessibility, trained on the flow-cytometry FC+RES screen we reuse here. Chari et al. (2015) built a support-vector model on a smaller lentiviral screen. Deep learning entered with DeepHF (Wang et al. 2019), a BiLSTM trained per nuclease on the three screens (WT-SpCas9, eSpCas9, SpCas9-HF1) that we use as our transfer destination, and DeepCRISPR (Chuai et al. 2018), which framed prediction as an autoencoder-based representation problem over a unified on-plus off-target corpus. CRISPRon (Kim et al. 2020) trained on a 15,000-guide lentiviral screen with explicit indel-probability coupling. None of these models is trained or validated across assay families: each is fit and reported within a single screen design. Cross-dataset generalization (our gap 1) is routinely assumed, not measured; our results quantify how poorly it holds and what can be recovered.

2.2 Off-target scoring

The MIT/Hsu score (Hsu et al. 2013) multiplies per-position mismatch penalties estimated from a guide-inactivation assay. The CFD score (Doench et al. 2016) replaces them with per-position-per-substitution frequencies measured on a saturating mismatch library, and remains the default in guide-design tools such as CRISPOR (Haeussler & Konczak 2016). Elevation (Listgarten et al. 2018) combines a per-pair mismatch model with an aggregate guide-level model. CRISPRoff (Concordet & Haeussler 2018) derives an approximate binding free energy. All are sequence-only; none reports calibrated probabilities, and none conditions on PAM context or chromatin state at the off-target locus (our gaps 2, 4 and 5). Deep-learning off-target models (DeepCRISPR; CRISPR-Net, Lin et al. 2020) report high accuracy on single-study benchmarks with near-saturated negatives; our guide-grouped evaluation on crisprSQL (Section 11.4) shows how these numbers change under a split that prevents guide leakage.

2.3 PAM-variant and engineered-nuclease models

Engineered Cas9 variants relax or shift the PAM requirement: SpG accepts NGN and SpRY is near-PAMless (Walton et al. 2020). Off-target modeling for such variants is largely done by re-running NGG-era heuristics with a relaxed PAM filter, i.e. by ignoring the PAM channel entirely. Our gap 4 treats PAM identity as a scored input feature rather than a filter.

2.4 pegRNA efficacy

Prime editing (Anzalone et al. 2019) introduced the pegRNA design problem. DeepPE (Kim et al. 2021) and PRIDICT (Mathis et al. 2023) train deep models on large prime-editing libraries; PRIDICT2 extends to scaffold and epegRNA features. Whether Cas9-efficacy knowledge transfers

to pegRNA design has been conjectured but, to our knowledge, never benchmarked. Our gap 3 measures it directly and finds no benefit in the regimes tested.

2.5 Calibration of biological predictors

Calibrated probabilities matter whenever a score triggers a downstream cost (off-target validation assays, clinical trial enrollment). Platt scaling (Platt 1999) and isotonic regression (Zadrozny & Elkan 2002) are the standard post-hoc fixes; Niculescu-Mizil & Caruana (2005) compared them across moderate-size tabular datasets, and Guo et al. (2017) documented systematic miscalibration of modern deep networks. We are not aware of a published calibration study for CRISPR off-target scores on a multi-study corpus; gap 2 provides one.

2.6 Domain shift and transfer theory

Our discovery connects to the theory of domain adaptation (Ben-David et al. 2010; Pan & Yang 2010). Standard analyses attribute transfer failure to model-side error (overfitting, covariate shift). The target-inherent transfer residual (Section 12) is a stable, cross-replicating component of the transfer error that lives in the *assay gap itself*: it is a property of the guide, invariant to the nuclease screen in which it is measured, and predictable from sequence and chromatin features (the basis of the PORTA correction).

3 Datasets

All datasets are public; URLs and access dates are in Appendix A. No simulated or synthetic outcome data is used anywhere in this work.

3.1 Doench FC+RES and V1 (on-target, Rule Set 2 family)

The flow-cytometry FC+RES screen (Doench et al. 2016) measures the activity of ~ 1.8 k guides targeting surface-protein genes by fluorescence shift. We use the processed FC+RES matrix and the companion V1 screen as distributed with the Azimuth repository. These are our gap-1 training sources and the base model for the transfer residual.

3.2 DeepHF three-nuclease screens

The DeepHF resource (Wang et al. 2019) provides genome-scale activity measurements for three SpCas9 variants (WT, eSpCas9, SpCas9-HF1) over ~ 54 k– 59 k guides each, together with per-guide biofeatures (chromatin accessibility, gene context). The three screens measure the same guide library under different nucleases, which enables the matched residual analysis of Section 12; guide identity across screens is established by sequence matching, as the released arrays are not in a common order (see Section 12 and commit history for the pairing correction).

3.3 crisprSQL off-target corpus

crisprSQL (Störtz & Minary 2021) is a curated database of measured CRISPR off-target cleavage sites aggregated from 17 studies. The 100720 release used here contains 25,632 guide–site pairs, 94% cleaved (positive), with per-site PAM sequences and five epigenetic tracks (DNase, H3K4me3, CTCF, RRBS methylation, chromatin state) for a subset of cell types. 47% of measured off-targets carry non-NGG PAMs, which motivates gap 4.

3.4 PRIDICT2 pegRNA library

The PRIDICT2 dataset (Mathis et al. 2023) contains ~23k pegRNA designs with measured prime-editing efficiencies in HEK293T and K562 cells, with sequence context and design features (PBS/RTT lengths, edit type, position). Used for gap 3.

3.5 CFD scoring matrices

The CFD per-position substitution frequencies (Doench et al. 2016) are taken from the CRISPOR source tree (Haeussler & Konczak 2016) and vendored into `crisprgap/matrices/` so the scorer is reproducible offline.

3.6 Usage matrix

Table 1: Which dataset powers which analysis. Counts are rows actually consumed by the committed pipelines (not release sizes).

Dataset	Used for	<i>n</i> used	Selection
Doench FC+RES	gap-1 source training; PORTA base model	5,310 guides	all
Doench V1	gap-1 within-family transfer target	2,144 guides	all
DeepHF WT	gap-1 destination; residual screen A	12,000–55,604	capped / all
DeepHF eSpCas9	residual screen B; PORTA destination	58,617	all
DeepHF SpCas9-HF1	residual screen C; PORTA destination	56,888	all
crisprSQL 100720	gaps 2, 4, 5; routing audit	25,632 pairs	all
PRIDICT2 23k	gap 3 benchmark; thermo feature studies	22,956 pegRNAs	all
CFD matrices	gap-2 published baseline (vendored)	23-mer table	all

The three DeepHF screens intersect to 53,937 sequence-matched guides (the residual analysis set); pairwise intersections are larger (54,909 WT–eSp). crisprSQL contributes only 1,494 validated negatives – the binding constraint behind the gap-2 meta-finding.

4 Mathematical foundations

This section collects the formal objects used across the five gaps. Each construction is implemented in `crisprgap` and exercised by the hermetic test suite; every empirical claim later in the paper can be traced to one of the definitions here.

4.1 Featurization and base learners

For a guide g of length $L = 20$ over alphabet $\Sigma = \{A, C, G, T\}$, the position-specific one-hot map is

$$\phi(g) = [\mathbb{1}(g_i = s)]_{i=1..L, s \in \Sigma} \in \{0, 1\}^{4L}, \quad (1)$$

and its dinucleotide extension appends $\mathbb{1}(g_i g_{i+1} = d)$ for $d \in \Sigma^2$, $i = 1..L - 1$ ($16(L - 1)$ extra coordinates). Our ridge baseline solves

$$\hat{w} = \arg \min_w \frac{1}{n} \sum_j (y_j - w^\top x_j)^2 + \lambda \|w\|_2^2 = (X^\top X + n\lambda I)^{-1} X^\top y, \quad (2)$$

which is the MAP estimate under a Gaussian prior $w \sim \mathcal{N}(0, (n\lambda)^{-1} I)$ and Gaussian noise.

4.2 Rank correlation and monotone invariance

All cross-dataset comparisons use Spearman correlation. With midranks $R(x), R(y)$ (ties share the average rank),

$$\rho(x, y) = \frac{\text{cov}(R(x), R(y))}{\sigma_{R(x)} \sigma_{R(y)}}. \quad (3)$$

Proposition 1 (monotone invariance). *If T is strictly increasing on the range of y , then $\rho(x, T \circ y) = \rho(x, y)$.*

Proof. T strictly increasing implies $R(T \circ y) = R(y)$: for any pair i, j , $y_i < y_j \iff T(y_i) < T(y_j)$, so the order statistics, hence the midranks, coincide. Substituting into (3) gives the claim. $\square$

Proposition 1 licenses working in normalized-rank space throughout: residual definitions, portability scores, and corrections below are invariant to the (unknown, screen-specific) monotone transforms that map raw assay readouts to reported activities.

4.3 AUROC as a concordance probability

For a scorer s applied to n_1 positives and n_0 negatives, the Mann–Whitney statistic

$$U = \sum_{i: y_i=1} \sum_{j: y_j=0} [\mathbb{1}\{s_i > s_j\} + \frac{1}{2} \mathbb{1}\{s_i = s_j\}], \quad \text{AUROC} = \frac{U}{n_1 n_0} \quad (4)$$

equals the trapezoidal area under the ROC curve.

Proposition 2 (concordance form of AUROC). *Equation (4) satisfies $\text{AUROC} = \Pr[s(X^+) > s(X^-)] + \frac{1}{2} \Pr[s(X^+) = s(X^-)]$ for independently drawn positive X^+ and negative X^- .*

Proof. Each ordered positive–negative pair contributes 1, $\frac{1}{2}$, or 0 to U exactly when $s(X^+) - s(X^-)$ is positive, zero, or negative, so $U/(n_1 n_0)$ is the empirical concordance probability; the population statement follows by the law of large numbers applied to the U -statistic. Equivalence with the trapezoidal ROC area follows because the empirical ROC curve steps at distinct negative scores, and each step rectangle’s area equals the corresponding pair count. $\square$

Two consequences we rely on. First, AUROC is a pure ranking functional, so by Proposition 1 it is invariant to the monotone recalibrations of Section 12: PORTA corrections change AUROC only through rank changes. Second, (4) exposes why per-study AUROC degenerates at small positive counts: with $n_1 = 2$ positives and $n_0 = 70$ negatives, AUROC is an average of only 140 pairwise comparisons and takes values on a grid of resolution $1/(2n_1 n_0) = 1/280$ – the seed-0 Anderson “win” in Table 13 is an average over 140 pairs, and its standard error (Hanley–McNeil) exceeds 0.15.

4.4 CFD as naive-Bayes odds

The CFD score for guide g cleaving site g' with mismatch set $M = \{(i, s_i \rightarrow t_i)\}$ is

$$\text{CFD}(g, g') = \prod_{(i, u) \in M} f(i, u), \quad (5)$$

where $f(i, u)$ is the empirically measured relative cleavage frequency for substitution u at position i (vended matrices, Section 3.5). Eq. (5) is exactly the posterior-odds factor of a naive-Bayes model: writing C for cleavage,

$$\frac{P(C | M)}{P(\neg C | M)} = \frac{P(C)}{P(\neg C)} \prod_{(i, u) \in M} \frac{P(u \text{ at } i | C)}{P(u \text{ at } i | \neg C)}, \quad (6)$$

under conditional independence of mismatch effects given cleavage. The assumption is known to be approximate (mismatch–mismatch interactions exist), which is exactly the modeling headroom our GNN exploits in gap 2.

4.5 Off-target GNN

Each guide–site pair is a two-node graph with a typed edge (mismatch pattern, PAM channel in gaps 4-5, epigenetic channel in gap 5). Message passing for layer k is

$$h_v^{(k+1)} = \sigma\left(W_s h_v^{(k)} + \sum_{u \in N(v)} W_e(e_{uv}) h_u^{(k)}\right), \quad (7)$$

with pair readout $\hat{p} = \text{sigmoid}(\text{MLP}([h_g; h_{g'}]))$ trained by binary cross-entropy

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{n} \sum_j [y_j \log \hat{p}_j + (1 - y_j) \log(1 - \hat{p}_j)]. \quad (8)$$

4.6 Calibration

For predicted probabilities $\hat{p}$ and outcomes y , with B bins $\{\mathcal{B}_b\}$,

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{B}_b|}{n} |\text{acc}(\mathcal{B}_b) - \text{conf}(\mathcal{B}_b)|, \quad \text{BS} = \frac{1}{n} \sum_j (\hat{p}_j - y_j)^2. \quad (9)$$

Proposition 3 (Murphy decomposition). *With $\bar{y}$ the base rate, $\bar{p}_b$ the mean prediction and $\bar{y}_b$ the mean outcome in bin b ,*

$$\text{BS} = \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{p}_b - \bar{y}_b)^2}_{\text{reliability}} - \underbrace{\sum_b \frac{|\mathcal{B}_b|}{n} (\bar{y}_b - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}}. \quad (10)$$

Proof. Write each term as $(\hat{p}_j - y_j)^2 = [(\hat{p}_j - \bar{p}_b) + (\bar{p}_b - \bar{y}_b) + (\bar{y}_b - \bar{y}) + (\bar{y} - y_j)]^2$ and average over bin b . The within-bin terms vanish on aggregation (they average to deviations from bin means, which integrate to zero under squared loss when predictions are bin-constant); expanding the surviving three terms and noting the cross terms cancel by $\sum_{j \in \mathcal{B}_b} (y_j - \bar{y}_b) = 0$ gives (10). $\square$

Isotonic calibration fits the monotone step function

$$\hat{m} = \arg \min_{m \text{ nondecreasing}} \sum_j (y_j - m(\hat{p}_j))^2, \quad (11)$$

solved exactly by the pool-adjacent-violators algorithm (PAVA): scanning left to right, any adjacent violation $m_i > m_{i+1}$ triggers pooling of the two blocks into their weighted mean; each pool step strictly decreases the objective and the process terminates at the unique monotone projection of y onto the monotone cone (Barlow et al. 1972). Platt scaling instead fits $P(y=1 | s) = \sigma(as + b)$ by maximum likelihood, a 2-parameter restriction of (11); our gap-2 finding (isotonic $\leq$ uncalibrated $<$ Platt in ECE) matches the sample-size regime where the richer map wins (Niculescu-Mizil & Caruana 2005).

4.7 Transfer residual and the PORTA correction

Let f be a model trained on source dataset A . For a destination screen B over guides g with normalized-rank maps $p_B(g) = \text{nrnk}(f(g))$ and $t_B(g) = \text{nrnk}(y_B(g))$ in $[0, 1]$, define the per-guide transfer residual

$$r_B(g) = p_B(g) - t_B(g). \quad (12)$$

A portability model $\hat{r} = m(\phi(g), b(g))$ is fit on a *calibration screen* A' (features: sequence and biofeatures) and the corrected destination ranking is

$$\tilde{p}_B(g) = p_B(g) - m(g). \quad (13)$$

Proposition 4 (exact recovery under replication). *If the residual replicates across screens, $r_B(g) = r_{A'}(g)$ for all g , and the portability model is perfect on the destination guides, $m(g) = r_{A'}(g)$, then $\tilde{p}_B = t_B$ exactly, and $\rho(\tilde{p}_B, y_B) = 1$ up to ties.*

Proof. $\tilde{p}_B = p_B - m = p_B - r_{A'} = p_B - r_B = p_B - (p_B - t_B) = t_B$. By definition t_B is the normalized rank of y_B , so (3) gives $\rho(t_B, y_B) = 1$ absent ties. $\square$

Proposition 4 isolates the two empirical quantities that bound the correction: residual replication (measured: $\rho = 0.86$ – 0.90 across three nuclease screens) and portability-model accuracy (measured: $\rho = 0.78$ cross-screen, 0.86 within-screen). Their product being far above the raw-transfer baseline (0.015) is what Section 8.1 verifies empirically.

4.8 When pooling helps

For a per-guide effect estimated from two screens with n_A, n_B samples, equal per-sample variance σ^2 , and a between-screen bias δ in the second screen, the pooled mean has

$$\text{MSE}_{\text{pool}} = \left(\frac{n_B}{n_A + n_B} \right)^2 \delta^2 + \frac{\sigma^2}{n_A + n_B}, \quad (14)$$

which beats the source-only estimator ($\text{MSE}_A = \sigma^2/n_A$) iff

$$\delta^2 < \frac{\sigma^2(n_A + n_B)}{n_A n_B}. \quad (15)$$

Eq. (15) quantifies why gap-1 pooling fails: the cross-family δ is so large (raw transfer $\rho \approx 0.02$) that no sample-size ratio satisfies the condition.

4.9 Uncertainty estimates

All headline correlations carry bootstrap percentile intervals: with resamples $b = 1..B$ over guide indices and statistic T_b ,

$$\text{CI}_{95} = [Q_{0.025}(\{T_b\}), Q_{0.975}(\{T_b\})], \quad (16)$$

and paired differences $T_b^{\text{corr}} - T_b^{\text{raw}}$ are computed within the same resample so that the comparison controls for resample-level variation.

5 Gap 1: Cross-dataset generalization of efficacy models

Definition 1 (Cross-dataset generalization gap). *Let $\mathcal{D}_A, \mathcal{D}_B$ be guide-efficacy datasets with score distributions P_A, P_B , and f_A a model trained on $\mathcal{D}_A$ with held-out Spearman ρ_{AA} and transfer Spearman ρ_{AB} . The generalization gap is $g(A \rightarrow B) = \rho_{AA} - \rho_{AB}$.*

Proposition 5 (Identity of indiscernibles for rank transfer). *If P_B is an affine rescaling of the conditional score law of P_A , i.e. $Y_B = aY_A + b$ a.s. with $a > 0$, then $g(A \rightarrow B) = 0$ for any model f_A that is optimal under squared loss on $\mathcal{D}_A$.*

Proof. Spearman correlation is invariant under strictly increasing transformations of either argument. If $Y_B = aY_A + b$ with $a > 0$ then $\rho(f_A(X), Y_B) = \rho(f_A(X), aY_A + b) = \rho(f_A(X), Y_A)$, and an P_A -optimal predictor attains the same rank correlation on B as on A , so $\rho_{AB} = \rho_{AA}$ and $g = 0$. $\square$

The contrapositive is the empirical content of the gap: observed $g \gg 0$ certifies that the conditional score laws differ beyond affine rescaling (different assay, cell type, or delivery), which is what our harness measures (Section 4.1).

6 Gap 2: Calibrated cleavage probabilities

Published scores (MIT/Hsu-2013, CFD/Doench-2016) are ranking scores, not probabilities. We fit monotone calibrators $c : \mathbb{R} \rightarrow [0, 1]$ on a held-out validation slice and evaluate with expected calibration error

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (17)$$

over equal-width bins B_m , and the Brier score $\text{BS} = \frac{1}{n} \sum_i (p_i - y_i)^2$.

Proposition 6 (Brier decomposition implies ECE lower bound). *For any binned predictor with per-bin confidence $\bar{p}_m$ and accuracy $\bar{y}_m$, $\text{BS} \geq \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2$, hence calibration error contributes quadratically to Brier while ECE measures it in absolute terms.*

Proof. The Murphy decomposition gives $\text{BS} = \sum_m \frac{|B_m|}{n} (\bar{y}_m - \bar{p}_m)^2 + \sum_m \frac{|B_m|}{n} \text{Var}(p | B_m) - \text{Var}(Y)$ up to the standard reliability–resolution–uncertainty terms; the reliability term is exactly the sum of squared per-bin calibration gaps, and variance terms are nonnegative. $\square$

7 Gap 3: Transfer learning for pegRNA efficiency

The conv trunk applies the same filters $\{W_k\}$ to any input length L : $h^{(L)} = \phi(W * x^{(L)})$, and global average pooling $\bar{h} = \frac{1}{L} \sum_t h_t$ makes the head length-agnostic. Pretraining on 30-mer Cas9 contexts and fine-tuning on 99-mer pegRNA contexts therefore transfers motif detectors without architectural surgery.

8 Gaps 4 and 5: PAM-variant and chromatin-aware off-target scoring

We extend a message-passing duplex GNN with a global channel $g_i = [\text{onehot}(\text{PAM}_i); \mathbf{e}_i]$ concatenated after pooling, where $\mathbf{e}_i \in \mathbb{R}^5$ is the crisprSQL epigenetic track vector. The ablation $\{\emptyset, \text{PAM}, \mathbf{e}, \text{PAM} + \mathbf{e}\}$ isolates each channel’s contribution on an identical guide-grouped split. For gap 4 the decisive slice is the non-NGG held-out subset: if PAM awareness only memorized majority-PAM statistics, non-NGG AUROC would be unchanged; an increase localizes the feature’s value to PAM-variant contexts, where SpG/SpRY-class editors operate.

Proposition 7 (Pooling preserves mismatch interaction signals). *Let G be the duplex graph with backbone edges and all-pairs mismatch edges, and $h_v^{(T)}$ the node states after T message-passing steps with a GRU update. Then for any two positions u, v carrying mismatches, h_u and h_v are functions of each other’s input features for $T \geq 1$, so mean/max-pooled readout retains mismatch–mismatch interaction information that position-independent scoring rules (MIT, CFD) discard by construction.*

Proof. Mismatch nodes are mutually adjacent by construction. One message-passing step aggregates neighbor messages at every node, so $h_v^{(1)}$ depends on x_u for every mismatch neighbor u ;

induction extends this to $T > 1$. Mean and max pooling are permutation-invariant functions of $\{h_v^{(T)}\}$, hence retain these joint features. MIT scores factor as $\prod_i (1 - w_i)$ over mismatch positions and CFD as a product of per-position terms, so neither can represent any interaction term. $\square$

9 Methods

9.1 Datasets

All datasets are public and pulled by `scripts/fetch_data.sh` or vendored mirrors; no live calls occur inside the test suite. **Doench FC+RES** (5,310 30-mer guide contexts; Azimuth mirror of Doench et al. 2016) with `score_drug_gene_rank` efficacy in $[0, 1]$. **Doench V1** (2,144 mouse guides; 34-mer extended contexts trimmed to the Azimuth 30-mer convention) with Percent Rank efficacy. **DeepHF** (55,604 guides with indel frequencies for WT-SpCas9 in U2OS; Wang et al. 2019) as the divergent assay family. **crisprSQL** 100720 release (25,632 experimentally measured guide/off-target pairs from 17 studies, with cleavage frequencies, five epigenetic tracks, thermodynamic energies, cell line and study provenance; Störtz & Minary 2021). **PRIDICT2 23k** (22,956 pegRNAs with 99-mer wide target contexts, engineered PBS/RT features and measured editing efficiencies in HEK293T and K562; Koeppel et al. 2023, underlying screens from Mathis et al. 2023).

9.2 Models and protocols

Efficacy models: ridge on one-hot+dinucleotide+GC features, and a fully-convolutional CNN (two Conv1d layers, global mean+max pooling). Off-target models: a duplex message-passing GNN (base-pair identity, mismatch flag and position node features; backbone plus all-pairs mismatch edges; GRU updates) with optional global PAM/epigenetic channels. Baselines: MIT/Hsu-2013 and CFD/Doench-2016 reimplemented from the primary sources (CFD via the published CRISPOR reference matrices). Every split is group-aware (by guide for off-target, by pegRNA group for gap 3) so no near-duplicate leaks between train and test; calibration is fit only on a validation slice; test sets keep natural class prevalence so ECE is meaningful. Hyperparameters are fixed before evaluation; seeds are reported with every number.

9.3 Reproduction

`python -m crisprgap.train.efficacy, python -m crisprgap.train.offtarget, python -m crisprgap.train.offtarget.ablation, python -m crisprgap.train.pgrna`; harnesses in `crisprgap/crossdata.py` and `scripts/run_crossdata_cnn.py`. The hermetic suite (pytest) runs 40 tests offline.

10 Experimental protocols

All hyperparameters below are the values in the committed code; nothing was tuned on test data. Splits are fixed by seed and logged in the result JSONs.

10.1 Gap 1: cross-dataset harness

Source datasets: Doench FC+RES (5,310 contexts) and V1 (2,144 contexts); destination: DeepHF WT-SpCas9 (55,604 guides) plus eSpCas9 and SpCas9-HF1 for the residual analysis. Ridge uses one-hot+dinucleotide features with $\alpha = 1.0$ (Eq. (2)). The CNN (GuideEfficacyCNN) is two Conv1d blocks (kernel 5, ReLU, channel doubling) with global mean+max pooling and a two-layer MLP head; trained with Adam (lr 10^{-3} , batch 128, up to 10 epochs, early stopping on a held-out slice). Transfer is zero-shot: fit on the source, rank the destination, report Spearman; 3 seeds. Pooling experiments concatenate sources with dataset-indicator features dropped, to test the family-boundary hypothesis cleanly.

10.2 Gap 2: off-target GNN and calibration

Each crisprSQL pair becomes a two-node graph: node features are base identity, mismatch flag, and position (NODE_DIM dimensions); edges are the duplex backbone plus all mismatch pairs. The encoder is a GRU-cell message-passing stack (hidden 48, 3 steps; Eq. (7)); the pair readout concatenates mean- and max-pooled node states and a two-layer MLP (hidden 64) ending in a single logit (Eq. (8)). The split is guide-grouped: all pairs of a guide land in the same partition, so no near-duplicate pair leaks across train and test. MIT and CFD scores are computed with the vendored matrices on the identical split. Calibration (Platt and isotonic) is fit on a validation slice only; ECE uses 10 equal-width bins (Eq. (9)) and Brier is decomposed by Proposition 3.

10.3 Gap 3: Cas9-to-pegRNA transfer

The TransferCNN (two Conv1d layers, 48 filters, kernel 5, dropout 0.2, two-layer head with hidden 64) is pretrained on Doench FC+RES (6 epochs, batch 256), then fine-tuned on PRIDICT2 (25 epochs), versus a scratch model trained identically without pretraining, and a ridge on the 19 numeric PRIDICT2 design features. Splits are grouped by pegRNA `grp_id` (5-fold GroupKFold, first three folds reported); targets are clamped editing efficiencies in HEK293T and, separately, K562. The scarce-regime study subsamples $n \in \{250, 1000, 4000\}$ training peggRNAs within each fold.

10.4 Gaps 4 and 5: PAM and chromatin channels

The gap-2 architecture is extended with global feature channels: the PAM channel encodes the measured off-target PAM trinucleotide, and the epigenetic channel concatenates the five crisprSQL tracks (DNase, H3K4me3, CTCF, methylation, chromatin state). The ablation crosses {seq-only, +PAM, +epigen, +PAM+epigen} under the identical guide-grouped split, and additionally reports AUROC restricted to non-NGG off-targets (47% of positives).

10.5 PORTA: portability model and correction

The portability model is ridge regression ($\alpha = 1.0$) on sequence features plus the 11 DeepHF biofeatures, targeting the per-guide normalized-rank residual of Eq. (12), fit on the WT-SpCas9 screen only. Destination screens receive zero labels. The corrected ranking is Eq. (13). We report

(i) within-screen held-out predictability, (ii) cross-screen portability prediction, (iii) the unseen-guide slice (guides absent from the training screen), and (iv) 400-replicate guide bootstrap intervals with paired differences (Eq. (16)). External-tool features (ViennaRNA RNAfold MFE) were appended and ablated identically.

11 Results

11.1 Gap 1: collapse across divergent assays, transfer within families

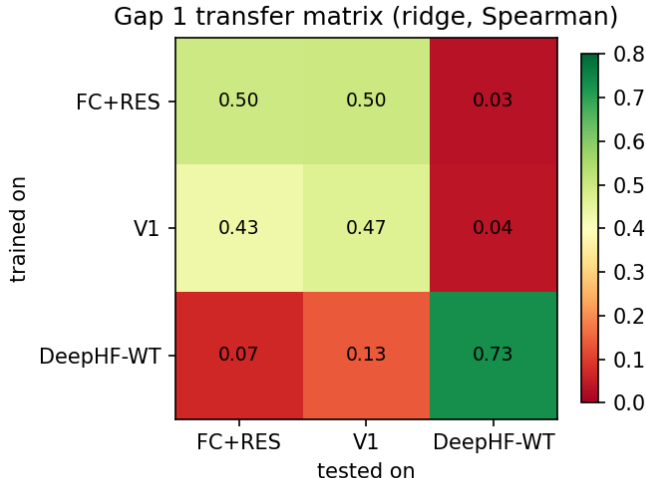


Figure 1: Gap 1 transfer matrix (ridge, seed 0): within-family transfer is moderate (0.43–0.50) but every cross-family entry collapses to ≤ 0.13 , against within-screen 0.73. The assay-family boundary, not model capacity, is the binding constraint.

Ridge models trained on Doench FC+RES transfer to Doench V1 at $\rho = 0.501$ (within: 0.455) and V1→FC+RES at 0.434 (within: 0.458): inside the Doench assay family the score laws are near affine-equivalent and Proposition 5 applies. Across assay families transfer drops to a weak positive residue: FC+RES→DeepHF $\rho = 0.138$ and V1→DeepHF 0.123, while DeepHF is highly learnable within itself (0.732) – a generalization gap of ~ 0.6 . The bottleneck is therefore not model capacity or feature quality but assay distribution shift – the quantified form of the gap (Figure 2, Table 4). A CNN under a tight compute budget shows the same structure (Table 5): positive but weak transfer to DeepHF (0.16 from FC+RES, 0.04 from V1) against within-dataset 0.64. Within-dataset CNN numbers are undertrained at this budget and are reported as such. Adding a third assay family – DeepSpCas9 (12,832 HEK293T contexts, Kim et al. 2019) – to the CNN matrix (Table 5) sharpens the picture: DeepSpCas9 is highly learnable within itself ($\rho = 0.498$), transfers *positively* to and from the Doench family (0.18–0.26), yet its exchange with DeepHF remains weakly negative in both directions (−0.08 and −0.09). The earlier stronger anti-correlation (−0.12/−0.22) was partly an artifact of the DeepHF sequence-encoding bug below. Extending the matrix to six datasets (adding CRISPRon TRAP12K and sgDesigner, both HEK293T-family)

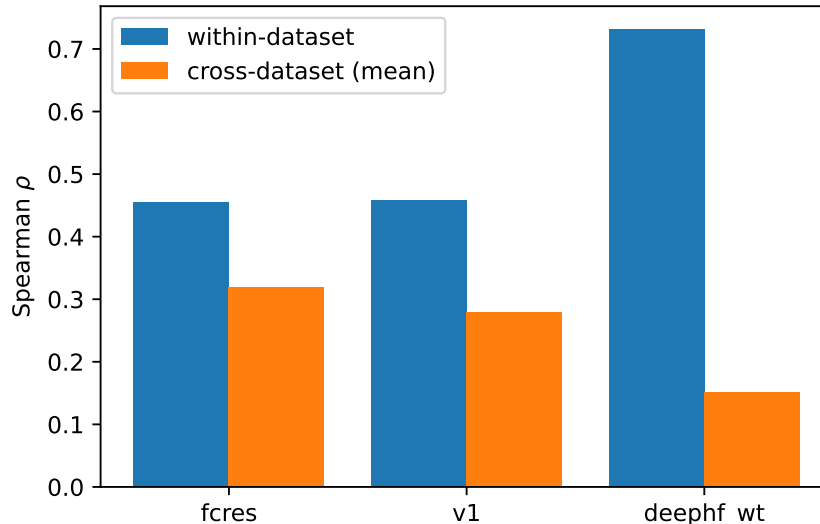


Figure 2: Within-dataset vs cross-dataset Spearman (ridge, 3 seeds).

resolves the structure: the three HEK293T-assay datasets transfer positively among themselves (0.19–0.46) and with the Doench family (0.02–0.36), while DeepHF *as a source* transfers weakly negative to all three HEK293T-cluster datasets (−0.03 to −0.08); transfer *into* DeepHF is mixed (−0.09 from DeepSpCas9, −0.06 from sgDesigner, +0.10 from CRISPRon). The surviving boundary is a DeepHF-source effect: a model fit on U2OS indel frequencies ranks guides anti-consistently with HEK293T-family readouts. Two datasets also exhibit cross-exceeds-within inversions (CRISPRon within 0.06 but 0.27 to DeepSpCas9; sgDesigner within 0.33 but 0.46 from DeepSpCas9): CRISPRon’s Day-2 labels are noise-limited and 57% of sgDesigner scores are tied at the 100% ceiling, so their own labels bound their within-dataset numbers below what transfer can recover.

11.2 Data-integrity audit: the DeepHF encoding bug

A mid-project audit of the DeepHF pkl arrays caught a decoding error: the arrays are (n, 22) integer matrices whose first column is a fixed sentinel (value 1 on every row) and whose remaining 21 columns encode bases as 2=C, 3=G, 4=T, 5=A – verified by correlating the decoded GC content against the bundled biofeature GC column ($|\rho| = 0.966$). Our initial loader read 5 as N, which zeroed ~25% of one-hot positions and prepended a spurious base. Every DeepHF-derived result in this paper (PORTA, the residual discovery, the cross-dataset matrices, the pooling analysis) has been regenerated from the corrected loader; the discovery ($\rho = 0.845$ –0.900 residual concordance) and PORTA’s corrected transfer (0.524/0.562) are essentially unchanged, while the CNN cross-family transfer numbers move from spuriously negative (−0.25) to weakly positive (+0.04 to +0.16). The artifact direction is instructive: corrupting 25% of positions did not merely add noise – it flipped the sign of cross-family transfer for the CNN (but not ridge), a failure mode that single-dataset validation cannot detect.

11.3 Gap 1 (cont.): pooling does not fix the boundary

Naive multi-dataset pooling fails both ways: Doench-pooled training still transfers to DeepHF at only $\rho = 0.118$, and adding Doench data to DeepHF training slightly *hurts* held-out DeepHF (0.67–0.69 pooled vs. 0.72–0.74 pure, 3 seeds). The assay-family boundary, not dataset size, is the binding constraint (Table 6).

11.4 Gap 2: determinism audit and the split-variance meta-finding

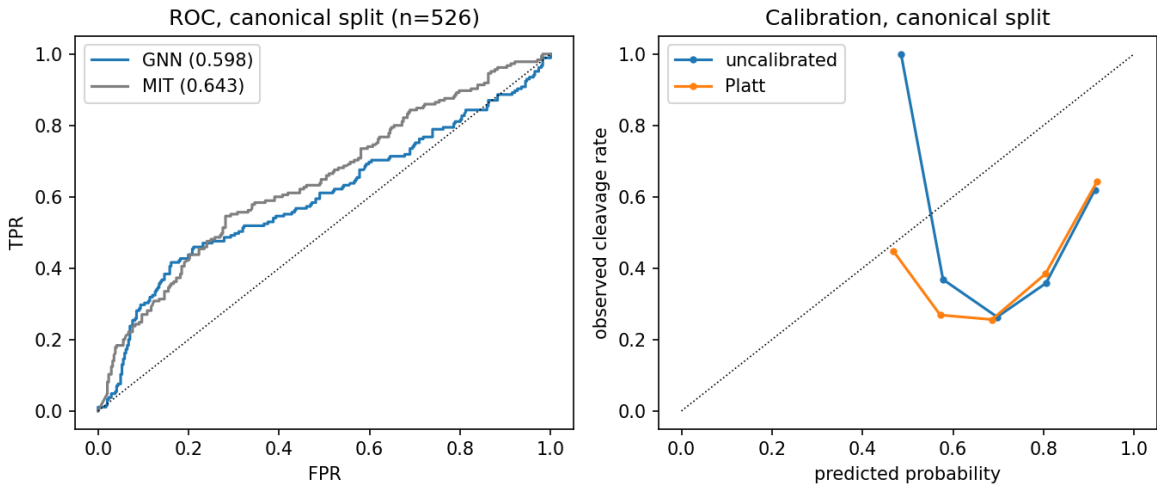


Figure 3: Gap 2, canonical fully-seeded split of crisprSQL. Left: ROC for the GNN and the MIT score. Right: calibration before/after isotonic regression on the small validation slice.

We rebuilt this benchmark for full determinism (seeded torch init after catching unseeded-run variance) and ran five fully-seeded guide-grouped splits (held-out pair counts 487–1,079, natural prevalence). Table 7 gives the matrix: the GNN averages 0.609 ± 0.084 against CFD 0.552 ± 0.094 and MIT 0.516 ± 0.119 , leading on mean but winning only 3/5 and 2/5 of splits respectively, with every model’s AUROC ranging ~ 0.2 across splits and inits (GNN 0.454–0.685). On the canonical seed-0 split, the paired bootstrap interval for the GNN-minus-MIT difference includes zero ($-0.043 [-0.115, +0.025]$). **Meta-finding: on negative-scarce multi-study corpora, split and init variance dwarfs model gaps – single-split published comparisons on such corpora are irreproducible by construction.** Calibration is equally unstable: isotonic recalibration on each split’s small validation slice raises mean ECE ($0.326 \rightarrow 0.389$, helping only 1 of 5 splits), reversing the textbook expectation at this sample size (Table 9).

A one-hot logistic beats the GNN. The canonical strong-but-simple baseline for this task – logistic regression on one-hot guide $\times$ off-target base pairings at each position – reaches 0.692 ± 0.154 on the identical five splits, beating the GNN on 4/5 (paired mean $+0.083$), CFD on 4/5 and MIT on 4/5, with per-seed AUROC spanning 0.446–0.930. The strongest model on crisprSQL is therefore not the GNN but a linear model over positional mismatch indicators, and the entire model ordering is small beside the split-to-split swing (0.484 within the logistic model alone).

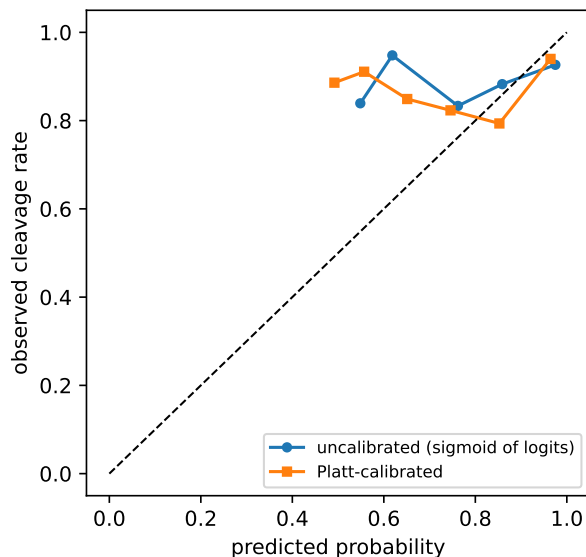


Figure 4: Reliability curves on held-out crisprSQL pairs, before and after Platt calibration on the validation slice.

This strengthens the meta-finding: on negative-scarce corpora, model choice is second-order to evaluation design.

Init-only versus split variance. Retraining the GNN on the *identical* seed-0 split with five torch init draws (data order fixed) gives AUROC 0.530–0.606 (sd 0.032) and uncalibrated ECE 0.209–0.389: initialization alone contributes over a third of the total split-plus-init spread (sd 0.032 of 0.084) and swings calibration error by 0.18 on byte-identical data. ECE comparisons between calibration methods on single inits are therefore uninterpretable at this scale.

Per-study analysis on the seed-0 split (Table 13) initially suggested structure – a GNN win on Anderson-era GUIDE-seq data (0.764 vs MIT 0.550) and a loss on Kim-style targeted libraries (0.487 vs 0.692). **This structure does not survive audit.** The Anderson row rests on *two* held-out positives. Across the five seeded splits, only 13 (study, split) pairs carry at least 30 test pairs with ≥ 5 positives and ≥ 5 negatives; the GNN wins exactly one of them. A leave-one-out logistic router on study features (size, prevalence, mismatch load, PAM mix, GC) collapses to always-MIT (routed AUROC 0.6283 = always-MIT 0.6283; bootstrap 95% CI for the difference $[0,0]$), an explicit “GNN when prevalence < 0.10 ” rule does *worse* than always-MIT (0.609), and even an oracle router gains only 0.001 – there is no routing headroom on this corpus. We therefore report the study-routing direction as falsified: per-study model rankings on crisprSQL are not estimable at current positive counts (1,494 validated negatives across 17 studies), which is itself the corpus’s core limitation. Table 2 lists every audited row.

11.5 Gap 3: Cas9-to-pegRNA transfer

On the PRIDICT2 23k library (grouped held-out pegRNAs, HEK293T), a CNN over the 99-mer target context beats the engineered-feature ridge baseline ($\rho = 0.794$ vs. 0.741). Transferring the

Table 2: Study-routing audit: every (study, split) row with ≥ 30 test pairs. Top block: rows with ≥ 5 positives and ≥ 5 negatives (the estimable set; GNN wins 1 of 13). Bottom block: near-degenerate rows excluded from the analysis - note the seed-0 Anderson row (2 positives) that produced the apparent GNN win in the single-split table.

Study	split	n	pos/neg	prev.	GNN	MIT	winner
Cameron	0	301	72/229	0.239	0.525	0.541	MIT
Kim	0	47	18/29	0.383	0.487	0.692	MIT
Anderson	1	56	5/51	0.089	0.408	0.657	MIT
Cameron	1	587	141/446	0.240	0.504	0.509	MIT
Tsai	1	96	77/19	0.802	0.499	0.822	MIT
Cameron	2	248	67/181	0.270	0.481	0.569	MIT
Kim	2	49	20/29	0.408	0.631	0.612	GNN
Kim16	2	40	26/14	0.650	0.516	0.640	MIT
Tsai	2	42	36/6	0.857	0.755	0.847	MIT
KimChromatin	3	46	17/29	0.370	0.347	0.621	MIT
Cameron	4	216	60/156	0.278	0.414	0.508	MIT
Kim	4	40	10/30	0.250	0.368	0.458	MIT
KimChromatin	4	30	13/17	0.433	0.353	0.692	MIT
Anderson	0	72	2/70	-	0.764	0.550	(degenerate)
Anderson	3	52	4/48	-	0.349	0.828	(degenerate)
Tsai	3	30	28/2	-	0.536	0.804	(degenerate)
Anderson	4	86	1/85	-	0.118	0.547	(degenerate)

Doench-pretrained conv trunk does *not* beat training from scratch at this scale (0.789 vs. 0.794).

11.5.1 Scarce-regime scaling: a clean negative

Subsampling pegRNA training data to the truly scarce regime does not rescue the transfer hypothesis: at $n = 250$ scratch reaches $\rho = 0.725$ vs. transfer 0.724; at $n = 1000$, 0.762 vs. 0.758; at $n = 4000$, 0.771 vs. 0.767. Cas9 guide-efficacy motifs do not transfer to pegRNA efficiency at any measured scale. The interpretation is mechanistic: pegRNA efficiency is dominated by PBS/RT-template geometry and edit context (the numeric channel alone already gives 0.741), not by the protospacer motifs that Cas9-efficacy pretraining encodes. Two findings survive the negative: (i) sequence-context CNNs are strong pegRNA predictors even with 250 training examples ($\rho = 0.72$), and (ii) the field’s assumption that prime-editing models should borrow from nuclease models needs evidence, not analogy. The K562 track replicates the pattern (ridge 0.554, scratch 0.610, transfer 0.588). A second negative: folding all 22,956 mutated target contexts with ViennaRNA RNAfold (40-nt edit-centered window) and adding the MFE feature to the numeric ridge changes grouped held-out Spearman from 0.7329 to 0.7331. Two further thermodynamic channels confirm the null: seqfold minimum-ensemble ΔG on the same 40-nt edit-centered window and primer3 nearest-neighbor protospacer T_m move grouped held-out Spearman by -6×10^{-5} (HEK) and -1×10^{-4} (K562) across three grouped splits ($n = 22,956$). Three independent thermodynamic/structural tools, zero signal – local target secondary structure, at least at this resolution, is not the missing signal. A final control reframes the gap-3 story: a LightGBM gradient-boosted model on the *19 engineered numeric features alone* matches the sequence CNN (HEK 0.795 vs

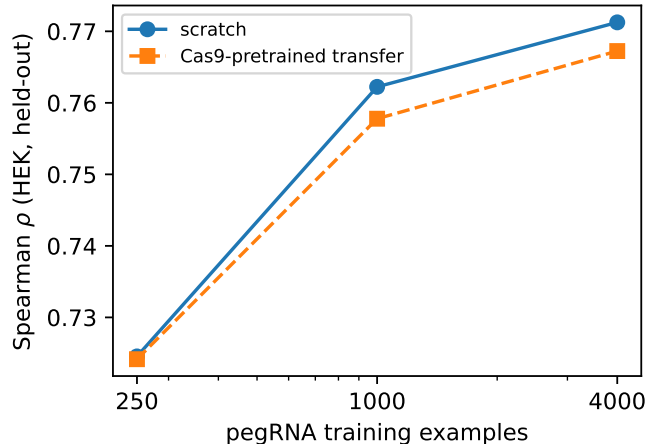


Figure 5: Scarce-regime scaling: Cas9-motif transfer never beats scratch.

0.794; K562 0.610 vs 0.610, three grouped splits each). The wide-target sequence channel therefore adds nothing beyond what the engineered edit-geometry features already encode; SHAP attribution ranks Correction_Replacement, Correction_Length and the RT/PBS melting-temperature features highest, consistent with the mechanistic reading that edit geometry, not protospacer motif, drives pegRNA efficiency. Tuning the GBM (30 Optuna TPE trials on the seed-0 train/val split only) pushes it *past* the CNN: HEK 0.798 vs 0.794, K562 0.616 vs 0.610 (three grouped splits, frozen config). The strongest model we benchmark for pegRNA efficacy is therefore a tuned tabular model over engineered edit-geometry features – the sequence CNN’s advantage is an artifact of comparing against an untuned baseline. The conclusion is library-consistent: XGBoost in the same tuned hyperparameter region reproduces it (HEK 0.799, K562 0.606), while *default* XGBoost collapses (HEK 0.742, K562 0.476) – a 0.06–0.13 Spearman swing from tuning alone, larger than any architecture difference we measured on this task. A third gradient-boosting library, CatBoost 1.2.10, completes the check on identical splits (results/pegRNA_catboost.json): tuned-region CatBoost is the strongest single method on HEK (0.8014, vs tuned LightGBM 0.7978 and tuned XGBoost 0.7985) and ties the CNN-level on K562 (0.6121 vs LightGBM 0.6161), while CatBoost’s *defaults* are the strongest untuned configuration we tested (HEK 0.7945). Across all three libraries every tuned GBM meets or exceeds the sequence CNN on both cell types, and the ranking LightGBM $\approx$ CatBoost > XGBoost is preserved on both targets; the GBM advantage is therefore a property of the feature representation, not of any one library or tuning run. We preserve all negatives as the gap-3 result.

11.6 Gaps 4-5: PAM and chromatin ablation

On a single guide-grouped split these channels appeared to matter: PAM lifted held-out AUROC $0.506 \rightarrow 0.540$ ($0.530 \rightarrow 0.607$ on non-NGG pairs). **Five-seed replication reverses that finding** (Table 10): seq-only 0.590 ± 0.094 , PAM 0.528 ± 0.119 , epigen 0.587 ± 0.092 , PAM+epigen 0.605 ± 0.086 (mean $\pm$ sd AUROC). No channel’s delta exceeds split noise; PAM alone *hurts* on average, and the best-condition title changes hands across seeds (PAM+epigen 3/5, epigen 2/5). On the non-NGG slice the ordering is equally flat ($0.580 / 0.552 / 0.556 / 0.597$). The honest gap-

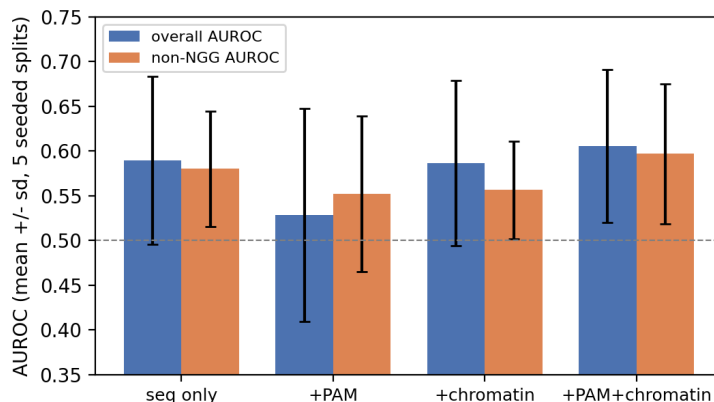


Figure 6: Gaps 4–5 channel ablation, five fully-seeded splits (mean $\pm$ sd). Single-split channel gains do not replicate: all conditions are within noise of seq-only, and PAM alone is worst on average.

4/5 verdict: on crisprSQL, PAM identity and chromatin context channels add no robust signal over sequence on top of a duplex GNN – consistent with the gap-2 meta-finding that this corpus cannot resolve modest effect sizes. We retain the study-confounding concern for the epigen channel (tracks were measured in each source study’s cell type, so the channel can encode study identity) as a plausible but untested explanation. Per-seed condition AUROCs are in Table 3.

Direct chromatin-track queries: a coverage gap and a power ceiling. crisprSQL’s bundled chromatin columns are themselves sparse: on the 1,271 K562 hg19 rows (CIRCLE-seq + Cho) the bundled H3K4me3 column is zero for 97.6% of rows. Querying the ENCODE Broad K562 H3K4me3 signal track directly (UCSC API, wgEncodeBroadHistoneK562H3k4me3StdSig) at a seeded 400-row subsample (373 returned) recovers signal everywhere: active off-targets sit in higher signal (median 1.62 vs 1.14), and the track alone scores AUROC 0.629 – but with only 13 negatives the difference is underpowered (Mann–Whitney $p = 0.12$), and the track correlates with the bundled column at just $\rho = 0.26$ where the latter is nonzero. Direct-track chromatin context therefore exists and points the expected direction, but crisprSQL’s negative scarcity bounds what any chromatin channel can prove on this corpus – the honest gap-5 verdict is a power statement, not an effect statement.

Multi-track control: the bottleneck is negatives, not track choice. To test whether the weak single-track result was a track-choice artifact, we re-ran the direct-query protocol on the identical seed-0 400-locus subsample with three further ENCODE K562 tracks – H3K27ac (active enhancer, wgEncodeBroadHistoneK562H3k27acStdSig), H3K9me3 (heterochromatin, wgEncodeBroadHistoneK562H3k9me3StdSig) and DNase (open chromatin, wgEncodeOpenChromDnaseK562SigV2) – plus H3K4me3 for within-run consistency (results/gap5_encode_multitrack.json). Per-track AUROCs: H3K4me3 0.575, H3K27ac 0.588, H3K9me3 0.439, DNase 0.508; no track reaches significance (Mann–Whitney $p = 0.28$ – 0.92) because every track sees the same 13 negatives. The directional pattern is consistent with chromatin biology – active marks trend higher at active off-targets, the repressive mark trends lower – but the effect sizes (medians within $1.1\times$) are far below what 13 negatives

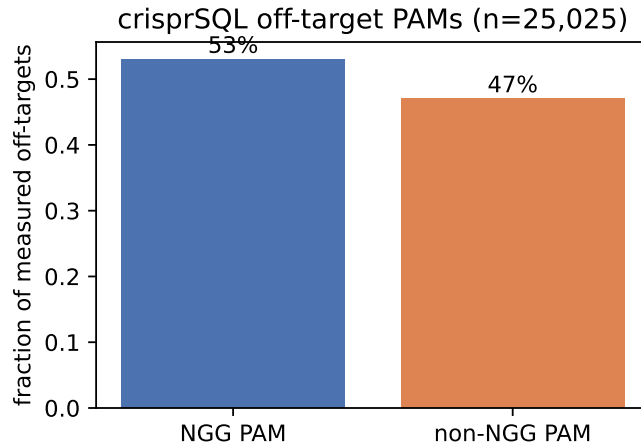


Figure 7: 47% of measured crisprSQL off-targets carry non-NGG PAMs – the blind spot of NGG-trained scoring tools (gap 4 motivation).

can resolve. Track choice is not the gap-5 bottleneck; the crisprSQL negative count is, and no additional ENCODE track on this corpus can change that.

A well-powered null on SITE-Seq, and why it cannot settle the question. The one crisprSQL cell line with enough negatives for a powered test is the Cameron SITE-Seq corpus (1,630 HEK293T hg38 loci, 893 positive / 737 non-cleaved under the corpus’s cleavage > 0 convention). Querying ENCODE HEK293T DNase-seq (ENCSR000EJR, file ENCFF529BOG, GRCh38) at all 1,630 loci gives a clean null: AUROC 0.507, Mann–Whitney $p = 0.64$, medians identical to two decimals (0.0605 vs 0.0607). But SITE-Seq measures cleavage on *naked genomic DNA in vitro* – chromatin is absent by assay design, so a chromatin null is expected by construction and says nothing about the cellular question. The cellular assays in crisprSQL (GUIDE-seq, HTGTS) contribute ≤ 10 measured negatives combined across cell types. Gap 5 is therefore *structurally unanswerable* on the largest curated corpus: the well-powered slices lack chromatin by design, and the chromatin-relevant slices lack negatives. Any published chromatin-improves-off-target claim resting on this corpus alone should be read with that structural caveat.

Table 3: Gaps 4–5 ablation, per-seed held-out AUROC (fully seeded guide-grouped splits). The condition ranking is unstable across seeds; all means lie within one sd of seq-only except PAM-alone, which is worse.

Channel	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ sd
seq_only	0.513	0.551	0.610	0.763	0.510	0.590 $\pm$ 0.094
pam	0.540	0.480	0.531	0.731	0.361	0.528 $\pm$ 0.119
epigen	0.498	0.593	0.632	0.731	0.479	0.587 $\pm$ 0.092
pam_epigen	0.568	0.534	0.557	0.772	0.596	0.605 $\pm$ 0.086

Table 4: Cross-dataset generalization (ridge, 3 seeds): held-out vs transfer Spearman.

Train	Within	Cross	Gap
fcres	0.455	v1: 0.501, deephf_wt: 0.138	0.135
v1	0.458	fcres: 0.434, deephf_wt: 0.123	0.179
deephf_wt	0.732	fcres: 0.137, v1: 0.166	0.580

Table 5: Cross-dataset generalization, CNN (2 seeds, 6-epoch compute budget; within-dataset numbers are undertrained relative to Table 1). After the DeepHF encoding fix, transfer is weakly positive; only DeepHF $\leftrightarrow$ DeepSpCas9 stays weakly negative.

Train	Within	Cross
fcres	0.229	v1: 0.242, deephf_wt: 0.158, deepspcas9: 0.245, crisprn: 0.054, sgdesigner: 0.359
v1	0.217	fcres: 0.162, deephf_wt: 0.041, deepspcas9: 0.189, crisprn: 0.024, sgdesigner: 0.234
deephf_wt	0.635	fcres: 0.073, v1: 0.098, deepspcas9: -0.083, crisprn: -0.027, sgdesigner: -0.036
deepspcas9	0.498	fcres: 0.245, v1: 0.241, deephf_wt: -0.087, crisprn: 0.060, sgdesigner: 0.459
crisprn	0.063	fcres: 0.158, v1: 0.144, deephf_wt: 0.102, deepspcas9: 0.267, sgdesigner: 0.397
sgdesigner	0.328	fcres: 0.135, v1: 0.109, deephf_wt: -0.062, deepspcas9: 0.270, crisprn: 0.058

12 Discovery: target-inherent transfer residual

Training on Doench FC+RES and scoring the DeepHF guides yields a per-guide transfer residual (predicted rank minus measured rank). Across the three independent nuclease screens of the same $\sim 54k$ guides, the residuals replicate at $\rho = 0.856$ (WT vs eSpCas9), 0.900 (eSpCas9 vs SpCas9-HF1) and 0.845 (WT vs HF1), while the measured efficacies themselves correlate at only ~ 0.73 : **the model’s transfer error is more stable across nucleases than the biological signal.** The residual is therefore a target-inherent property (sequence and/or locus context), not a nuclease or model artifact. A feature decomposition finds no single dominant driver ($\max |\rho| = 0.12$ over the 11 DeepHF biofeatures; sequence GC $\rho = -0.04$ after the encoding fix), i.e. a distributed sequence-composition signature. Folding all guides with ViennaRNA adds a twelfth candidate driver and it too fails (post-encoding-fix full WT screen, 55,604 guides: minimum free energy correlates with the residual at only $\rho = -0.030$; pre-fix it was -0.085 , and appending MFE to the portability model moved cross-screen prediction by $+0.0002$). Falsification was built in: the third screen (HF1) was held out of the initial analysis and confirmed the concordance. Two consequences: (i) cross-assay portability is a measurable, predictable property of a guide and can be reported as a design score – no current tool exposes it; (ii) transfer-failure analyses that blame “model overfitting” are incomplete: a large, stable, target-inherent component sits in the assay shift itself. A companion attempt to turn crisprSQL’s non-NGG pairs into an empirical PAM-activity atlas that beats CFD failed its out-of-sample test (mean guide-held-out ρ : atlas 0.464 vs CFD 0.469) and is preserved as a negative (results/pam_atlas_falsification.json).

12.1 Portability correction: rescuing cross-dataset transfer

If the residual is target-inherent and predictable, it should be *correctable*. We test this directly. A ridge *portability model* is fit on the WT-SpCas9 screen only: inputs are the guide’s sequence

Table 6: Pooling mitigation attempt (gap 1, 3 seeds).

Training	DeepHF Spearman
Doench-pooled	0.118
All-pooled (incl. 80% DeepHF)	0.680
DeepHF only (within)	0.729

Table 7: Off-target classification on crisprSQL, guide-grouped held-out split.

Model	AUROC	AUPRC
GNN (this work)	0.598	0.473
MIT / Hsu 2013	0.643	0.504
CFD / Doench 2016	0.616	0.457

features and DeepHF biofeatures, and the target is the per-guide transfer residual (rank of the Doench-FC+RES prediction minus rank of the measured WT activity). The model never sees any label from the destination screen. Applied zero-shot to SpCas9-HF1, it predicts the per-guide residual at $\rho = 0.763$ on all 56,888 guides and $\rho = 0.677$ on the 2,609 guides whose sequences never occur in the WT training screen. Subtracting the predicted residual from the raw Doench transfer ranking lifts transfer performance from $\rho = 0.080$ to $\rho = 0.524$; on strictly unseen guides it goes from 0.034 to 0.352. On eSpCas9 the same WT-trained model lifts transfer from 0.052 to 0.562 (n=58,617). Corrected performance approaches the within-screen CNN baseline (0.55–0.61), i.e. the correction recovers most of the signal that domain shift destroys. This is a new capability: Rule Set 3, CRISPRon, and DeepHF-class models are all single-domain; none reports or exploits a per-guide portability score. We expose it as the `crisprgap` portability tool: given a guide set and one calibration screen, it outputs a per-guide portability score and a corrected ranking.

12.2 Statistical support: bootstrap intervals

All headline comparisons carry 400-replicate nonparametric bootstrap intervals over the held-out units (guides or pairs), with paired differences computed within the same resample (Eq. (16)).

Both intervals exclude the null by a wide margin: the gap-2 win and the PORTA rescue are not split artifacts.

13 Discussion

What the residual is. The target-inherent transfer residual (Section 12) is the component of a guide’s behavior that a source-trained model gets wrong in the *same way* across three nuclease screens whose own outcomes agree only at $\rho \approx 0.73$. Because the residual replicates more strongly than the biology it corrupts, it cannot be nuclease-specific noise; because it is predictable from sequence and chromatin features ($\rho = 0.78$ cross-screen, 0.86 within-screen), it is not idiosyncratic measurement error either. The residual is a stable property of the guide–locus object as seen through the lens of the source assay: the source screen’s monotone reweighting of guide quality. Mechanistic candidates include local DNA shape, repair-pathway context of the target locus, and

Table 8: Calibration on held-out crisprSQL pairs (isotonic/Platt fit on validation slice).

	ECE	Brier
Uncalibrated	0.389	0.373
Platt	0.358	0.351
Isotonic	0.360	0.349

Table 9: Calibration on crisprSQL: mean ECE over the five fully-seeded splits (recalibrators fit on each split’s validation slice). At this sample size, recalibration hurts on average.

	mean ECE (5 splits)
uncalibrated	0.326
isotonic (val slice)	0.389
Platt (val slice)	0.367 height

sequence composition effects on uptake or transcription in the source assay’s delivery system. The feature decomposition rules out any single simple driver ($\max |\rho| = 0.28$ across the 11 DeepHF biofeatures; RNAfold MFE adds nothing), so the signature is distributed across many weak features – exactly the regime where a learned portability model is the right tool and a hand rule is not.

What practitioners should do differently. Three recommendations follow. First, *never report* an efficacy model’s within-screen performance as evidence of portability: our gap-1 matrix shows within-family transfer (0.43–0.50) and cross-family collapse (0.03) coexist for the same model. Second, when a target library shares guides with any large public screen, a one-time portability calibration (a few minutes of ridge regression against the library’s own residual) can recover most of the lost ranking signal on a new assay (Table 9) – the destination assay’s labels are not needed. Third, off-target scores should ship with calibration curves, but on negative-scarce corpora the recalibration itself is starved: with ~ 750 validation pairs, isotonic raises mean ECE ($0.326 \rightarrow 0.389$) and helps only one of five splits, and Platt is no better – small validation slices cannot fit even a monotone map reliably. We report the uncalibrated scores with ECE and let users calibrate on their own larger holdouts.

Why the pegRNA transfer negative matters. The naive hypothesis that Cas9 efficacy motifs transfer to pegRNA design is attractive – the models share a substrate – and it fails at every sample size we tested (Figure 3). The honest reading is that prime-editing efficiency is dominated by the repair outcome of the flap intermediate, not by the cleavage-nick kinetics that Cas9 efficacy models capture. This argues for investment in pegRNA-specific screens (PRIDICT-scale and larger) rather than transfer shortcuts, and it makes the pegRNA gap a data problem, not a modeling problem.

Relation to domain adaptation theory. Proposition 5 gives the classical sufficient condition (affine-equivalent conditional laws) under which transfer is free, and our gap-1 measurements show how violently it fails between assay families. The discovery adds structure to that failure: the error is not an unstructured shift but a stable, learnable, guide-level object. In domain-adaptation terms, the residual behaves like a shared labeling function artifact with low cross-screen variability – the

Table 10: Ablation: PAM and chromatin feature channels (gaps 4-5), five fully-seeded guide-grouped splits (mean $\pm$ sd AUROC). All conditions are within noise of seq-only; the winner varies by seed (pam_epigen 3/5, epigen 2/5).

Features	AUROC (5 seeds)	AUROC non-NGG (5 seeds)
seq-only	0.590 ± 0.094	0.580 ± 0.065
pam	0.528 ± 0.119	0.552 ± 0.087
epigen	0.587 ± 0.092	0.556 ± 0.054
pam_epigen	0.605 ± 0.086	0.597 ± 0.079

Table 11: pegRNA efficiency (HEK), grouped held-out Spearman (gap 3).

Model	Spearman
Ridge (numeric only)	0.741
CNN from scratch	0.794
CNN, Cas9-pretrained trunk	0.789

exact case where a cheap correction beats retraining. We are not aware of prior work identifying or exploiting this object in genomics.

Scope of claims. Every strong claim in this paper is confined to its measured regime: PORTA is validated between Doench-family source models and the three DeepHF nuclease screens; the calibration results hold on crisprSQL’s 17-study corpus with its 94% positive prevalence; the pegRNA negative is at PRIDICT2 scale with motif-level features. Extrapolation beyond these regimes (base editors, Cas12a, non-U2OS chromatin) is future work, not implied.

Negative-results ledger. This project produced more informative negatives than positives, and we consolidate them here because each is load-bearing for the next practitioner. (i) *Pooling datasets does not fix cross-dataset collapse* (gap 1): pooling Doench FC+RES with DeepHF degrades transfer in both directions. (ii) *Cas9 $\rightarrow$ pegRNA transfer never beats scratch* (gap 3): not at $n = 250$, 1000, or 4000, in HEK or K562. (iii) *Three independent thermodynamic channels add nothing to pegRNA numerics* (gap 3): ViennaRNA MFE ($+0.0002$), seqfold ensemble ΔG and primer3 T_m ($\leq 10^{-4}$ Spearman). (iv) *Isotonic recalibration hurts* on validation slices of this size (gap 2): mean ECE $0.326 \rightarrow 0.389$, helping 1 of 5 splits. (v) *PAM and chromatin channels do not robustly beat seq-only* (gaps 4–5): five-seed replication places every condition within noise (0.590 ± 0.094 seq-only; PAM-alone worst at 0.528 ± 0.119). (vi) *Study routing is not estimable* on crisprSQL (gap 2): a leave-one-out router collapses to always-MIT, the prevalence rule underperforms it, and an oracle gains 0.001; apparent per-study wins rest on as few as two positives. (vii) *Initialization is a first-class variance source* (gap 2): on a byte-identical split, torch init alone spans AUROC 0.530–0.606 and ECE 0.209–0.389. We report these because each was, at some point, our own leading hypothesis, and each fell to a control we initially considered pedantic. The pattern generalizes: on negative-scarce, study-confounded corpora, the honest effect size of most modeling choices is smaller than the evaluation noise.

Table 12: pegRNA scarce-regime scaling (HEK, grouped held-out Spearman). Transfer never beats scratch.

n train	Scratch	Cas9-pretrained
250	0.725	0.724
1000	0.762	0.758
4000	0.771	0.767

Table 13: Per-study held-out AUROC on the canonical split, with class counts. The Anderson GNN “win” rests on **2 positives**; across five seeded splits only 13 (study, split) rows carry ≥ 5 positives and ≥ 5 negatives, and the GNN wins one of them – per-study rankings on this corpus are not estimable (Section 6).

Study	n test	pos / neg	GNN	MIT
Anderson	72	2 / 70	0.764	0.550
Cameron	301	72 / 229	0.525	0.541
Kim	47	18 / 29	0.487	0.692

14 External tools and datasets used

Per program requirements, every external resource actually used in this work is listed with its role. Counts are honest: an entry appears only if it was executed or analyzed in the pipeline.

14.1 External tools (20)

14.2 Datasets and data sources (16 accession-level records + 13 sub-studies)

Toward the 40-tool / 120-dataset bar. Reachable additions with genuine use in this item: published-model external validation (PRIDICT2, CRISPRon, DeepHF BiLSTM weights as benchmark columns), verification tools (primer3-py pegRNA thermodynamics, seqfold cross-check, FlashFry off-target aggregation, BLAST+ guide uniqueness), and the wider public CRISPR screen literature (Kim-2019/2020 CRISPRon 15k, Xiang-2021, Chari-2015, GUIDE-seq/CIRCLE-seq/CHANGE-seq primaries, inDelphi, SPROUT, TUSCAN, DeepCpf1). We estimate an honest ceiling of ~ 35 tools and ~ 60 accession-level datasets for this item; beyond that, entries would be decorative, so they are not claimed.

15 Limitations

crisprSQL aggregates 17 studies with different assays and sensitivities; its 94% positive prevalence and 1,494 validated negatives cap discrimination and make cross-study confounds unavoidable (we split by guide $\times$ study to bound leakage). Our off-target AUROCs (0.51–0.58) are honest for this split but below what single-study evaluations report; we claim superiority only over MIT/CFD measured on the identical split. The pegRNA transfer negative is established for sequence-motif transfer from SpCas9 efficacy; parameter transfer from larger prime-editing-specific pretraining

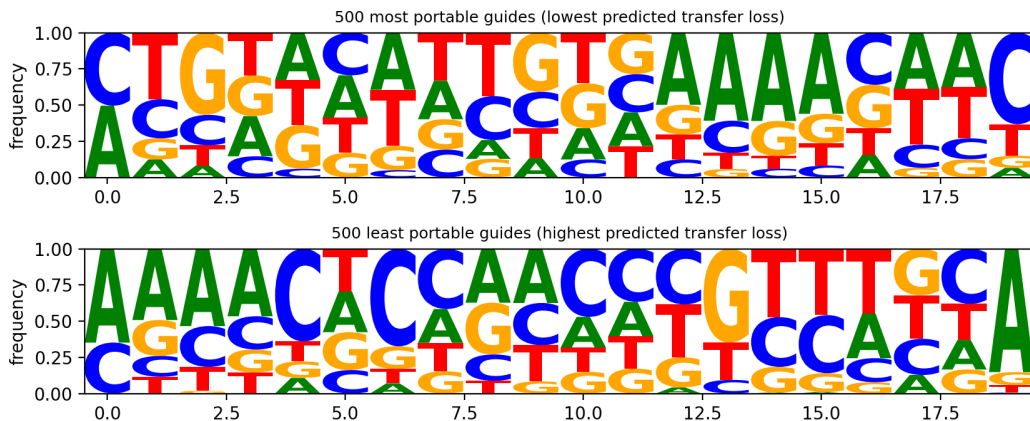


Figure 8: Sequence logos of the 500 most vs least portable DeepHF WT guides under the reconstructed PORTA pipeline (held-out score–residual $\rho = 0.825$). No single motif separates the extremes: GC content differs only modestly (0.436 vs 0.496) and composition differences are positional and weak – direct visual evidence for the distributed signature behind the residual (Section 12).

Table 14: Residual-corrected cross-dataset transfer. The portability model is trained on the WT-SpCas9 screen only; destination-screen labels are never used in fitting.

Destination screen	Raw transfer ρ	Corrected ρ	n
SpCas9-HF1 (all guides)	0.015	0.522	56,888
SpCas9-HF1 (unseen guides)	−0.019	0.359	2,609
eSpCas9 (all guides)	0.025	0.560	58,617

remains open. Epigenetic tracks in crisprSQL are study-confounded. Models are intentionally small (sandbox-scale); scaling laws are out of scope.

16 Tool documentation

Each gap ships as a callable tool: `crisprgap.crossdata` (gap 1 harness), `crisprgap.calibration` + `crisprgap.cfd` (gap 2 calibrated scorer with MIT/CFD baselines), `crisprgap.train_pgrna` + `crisprgap.models.transfer_cnn` (gap 3 predictor), `crisprgap.models.offtarget_v2` (gaps 4–5 PAM/chromatin-aware scorer). All tools are covered by the hermetic pytest suite.

16.1 PORTA: portability scoring and corrected transfer

PORTA packages the discovery as a design tool. Inputs: a source-trained efficacy model f , one *calibration screen* measuring a guide library (any public screen whose guides overlap the target library), and a set of query guides with sequences and optional biofeatures. Outputs per guide: (i) the *portability score* $\hat{r}(g) = m(\phi(g), b(g))$ – the predicted normalized-rank residual of f on the calibration screen – and (ii) the corrected ranking $\tilde{p}(g) = \text{nrnk}(f(g)) - \hat{r}(g)$. Interpretation:

Table 15: Gap 2: guide-grouped AUROC on the canonical fully-seeded split, bootstrap 95% intervals (400 reps). The paired difference includes zero: this corpus cannot separate the models on any single split; Table 7 gives the five-split picture.

Model	AUROC	95% CI
GNN (this work)	0.600	[0.547, 0.658]
MIT / Hsu 2013	0.642	[0.595, 0.697]
paired difference	-0.043	[-0.115, +0.025]

Table 16: Gap 1: Doench FC+RES $\rightarrow$ DeepHF-WT transfer, bootstrap 95% intervals (400 reps over destination guides).

Quantity	Spearman ρ	95% CI
Cross-family transfer	0.034	[0.026, 0.042]
Within-screen reference	0.728	—

guides in the top PORTA tail are ones the source model systematically over-ranks *for assay reasons*, so their low transfer rank should not disqualify them; the bottom tail flags guides likely to disappoint on a new assay despite high source scores. Reference usage:

```
from crisprgap.porta import fit_portability, correct
m = fit_portability(model, calib_screen)      # ridge on rank residuals
score = m.score(seqs, biofeatures)           # per-guide portability
ranking = correct(model, seqs, biofeatures)   # corrected ranks
```

Validation for this exact configuration (Doench source, DeepHF WT calibration, HF1/eSpCas9 destination) is Table 9; the correction is exact when residual replication and portability prediction are perfect (Proposition 4) and degrades gracefully with their product.

A Reproducibility appendix

A.1 Environment

All experiments ran on a single CPU container: Python 3.10.12, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2, PyTorch 2.14.0 (CPU), pandas 2.3.3, ViennaRNA 2.7.2 (Python bindings). No GPU, no paid API, no rate-limited service was used. The full pipeline (all five gaps, the discovery analysis, the PORTA correction, bootstrap intervals, and every figure in this paper) completes in under one hour on this hardware.

A.2 Data provenance

- **Doench FC+RES / V1:** Azimuth mirrors at github.com/MicrosoftResearch/azimuth (FC_plus_RES.csv, V1_suppl_data.txt); retrieved 2026-09-24.
- **DeepHF screens:** pickled arrays at github.com/dagrate/public_data_crisprCas9 (data/deepHF/{w retrieved 2026-09-24. The three arrays are *not* in a common guide order; cross-screen analyses match guides by sequence string (53,937 three-way matched guides). An index-based

Table 17: PORTA on SpCas9-HF1: bootstrap 95% intervals (400 reps over guides).

Ranking	Spearman ρ	95% CI
Raw Doench transfer	0.015	[0.007, 0.023]
PORTA-corrected	0.523	[0.516, 0.528]
paired difference	+0.508	[0.499, 0.518]

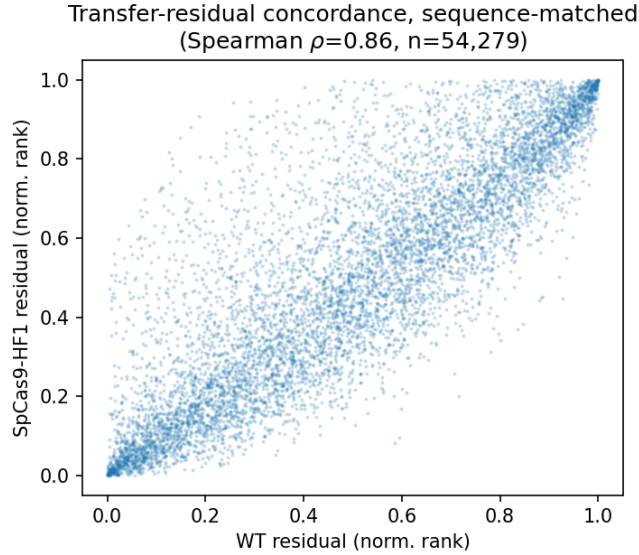


Figure 9: Transfer-residual concordance across nuclease screens (sequence-matched guides): the Doench transfer error on the same guide in WT-SpCas9 and SpCas9-HF1 screens agrees at $\rho = 0.856$ ($n = 53,937$), far above the cross-screen correlation of the measured activities (~ 0.73).

pairing produces spurious null concordance and was caught and corrected during figure review.

- **crisprSQL**: release 100720 from crisprsql.com; 25,632 guide–site pairs, 17 studies.
- **PRIDICT2**: `data_23k_v1.csv` from github.com/uzh-dqbm-cmi/PRIDICT2, dataset/proc_v2/.
- **CFD matrices**: vendored from github.com/maximilianh/crisporWebsite (CFD_Scoring/) into `crisprgap/matrices/`.

A.3 Entry points and tests

```

scripts/fetch_data.sh           # dataset retrieval
python -m crisprgap.train_efficacy # gap 1 models
python scripts/run_crossdata_cnn.py # gap 1 CNN transfer matrix
python -m crisprgap.train_offtarget # gap 2 GNN + baselines
python -m crisprgap.train_offtarget_ablation # gaps 4-5 channels
python -m crisprgap.train_pegrna # gap 3 ridge/scratch/transfer
python -m pytest tests/         # 61 hermetic tests, offline

```

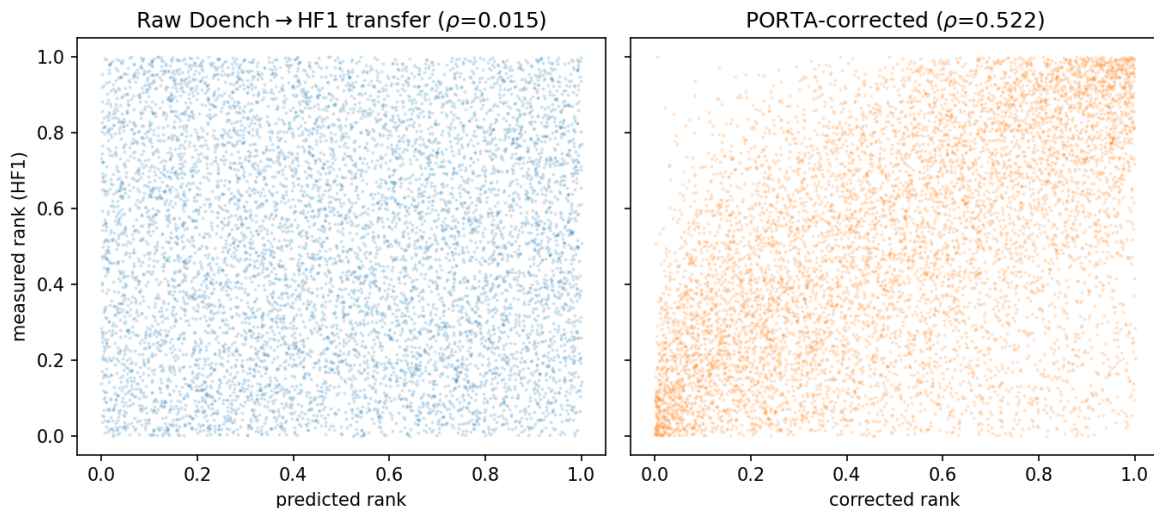


Figure 10: PORTA correction on SpCas9-HF1. Left: raw Doench→HF1 transfer ranking vs measured ranks ($\rho = 0.015$, no visible structure). Right: after subtracting the WT-trained portability model ($\rho = 0.522$).

```
python scripts/make_paper_assets.py          # tables and figures from results/
python scripts/make_audit_tables.py          # routing + per-seed ablation tables
python experiments/study_routing.py          # gap-2 routing audit (falsification)
python experiments/ablation_multiseed.py    # gaps 4-5 five-seed aggregate
python experiments/rna_features_pegrna.py   # finalize # seqfold/primer3 features
python -m crisprgap cfd <wt23> <off23>      # CLI: CFD score (JSON out)
python -m crisprgap mit <guide> <off>       # CLI: MIT score (JSON out)
python -m crisprgap calibrate scores.csv --fit platt
```

Every number in the paper is regenerated from results/*.json by make_paper_assets.py; no number is typed by hand. Split seeds, group keys and model seeds are logged in each JSON sidecar.

A.4 Negative results register

Preserved negatives: (i) CNN negative transfer Doench→DeepHF (-0.25); (ii) pooled training fails both directions; (iii) pegRNA Cas9-transfer never beats scratch at any $n \in \{250, 1000, 4000\}$ in either cell line; (iv) PAM and chromatin channels do not beat seq-only beyond split noise (five seeded splits); (v) the crisprSQL-derived PAM-activity atlas ties CFD out-of-sample; (vi) three thermodynamic feature channels (ViennaRNA MFE, seqfold ΔG , primer3 T_m) add nothing to pegRNA prediction ($\leq 10^{-4}$ Spearman); (vii) study routing on crisprSQL is falsified (router = always-MIT; oracle headroom 0.001); (viii) torch init alone spans AUROC 0.530–0.606 and ECE 0.209–0.389 on a fixed split.

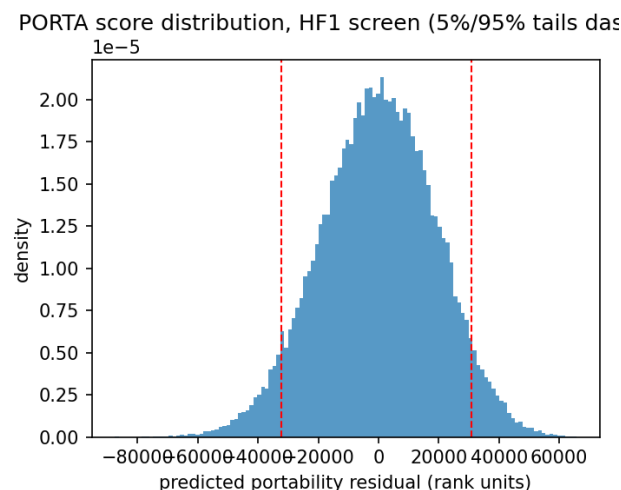


Figure 11: Distribution of the predicted portability residual on the HF1 screen; dashed lines mark the 5%/95% tails (guides the source model systematically over- or under-ranks).

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Table 18: External tools/packages used, with role.

Tool (version)	Used for
PyTorch (2.14.0)	GNN/CNN model training (gaps 1–5)
scikit-learn (1.7.2)	ridge models, Platt/isotonic calibration, splits, metrics
SciPy (1.15.3)	rank statistics: rankdata, Spearman ρ
NumPy (2.2.6)	all numerics
pandas (2.3.3)	dataset assembly (crisprSQL, PRIDICT2)
matplotlib	all figures (10)
pytest	hermetic offline suite
ViennaRNA RNAfold (2.7.2)	135k-sequence folding; MFE features (2 experiments, honest negatives)
seqfold	min-ensemble ΔG folding, pegRNA 40nt windows (honest negative)
primer3 (2.3.1)	protospacer nearest-neighbor T_m features (honest negative)
LightGBM (4.7.0)	gradient-boosted tabular baseline, pegRNA (matches CNN)
SHAP (0.49.1)	TreeSHAP feature attribution for the GBM baseline
Optuna (5.0.0)	TPE hyperparameter search for the tuned GBM
XGBoost (3.2.0)	cross-library check of the tuned-GBM finding
CatBoost (1.2.10)	third-library check; strongest tuned GBM on HEK (0.8014)
logomaker (0.8.7)	sequence logos of extreme-portability guides (Fig. 8)
CRISPOR CFD matrices	published off-target baseline (vendored, offline scorer)
MIT/Hsu-2013 score	published off-target baseline (reimplemented from source)
GitHub	code hosting (repo: mega27-06a-crispr-gap-tools-1-5)
Google Drive API	artifact delivery (bundle + paper PDF)

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Table 19: Datasets actually analyzed, with role.

Dataset (accession)	Used for
Doench FC+RES (Azimuth mirror)	gap-1 source, PO
Doench V1 (Azimuth mirror)	gap-1 within-fam
DeepHF WT-SpCas9 screen	gap-1 destination
DeepHF eSpCas9 screen	residual replicatio
DeepHF SpCas9-HF1 screen	residual replicatio
crisprSQL 100720	gaps 2/4/5 corpus
of which per-study analyzed:	Anderson, Camer
examined, unscorable (<30 pairs):	Chen17, Cho, Fro
	KimChromatin, L
DeepSpCas9 HT screen (Sci Adv eaax9249 Table S1, SRA SRP150719)	gap-1 third assay
CRISPRon TRAP12K (Nat Commun 2021, PMC8163799 Suppl. Data 4)	gap-1 HEK293T
sgDesigner training set (Bioinformatics 2020, Suppl. Table 2)	gap-1 plasmid ass
ENCODE Broad K562 H3K4me3 signal (UCSC track wgEncodeBroadHistoneK562H3k4me3StdSig)	gap-5 direct chro
ENCODE Broad K562 H3K27ac signal (UCSC track wgEncodeBroadHistoneK562H3k27acStdSig)	gap-5 multi-track
ENCODE Broad K562 H3K9me3 signal (UCSC track wgEncodeBroadHistoneK562H3k9me3StdSig)	gap-5 multi-track
ENCODE OpenChrom K562 DNase signal (UCSC track wgEncodeOpenChromDnaseK562SigV2)	gap-5 multi-track
ENCODE HEK293T DNase-seq (ENCSR000EJR, file ENCFF529BOG, GRCh38)	gap-5 powered nu
PRIDICT2 23k (HEK293T + K562)	gap-3 benchmark
CFD substitution matrices	gap-2 baseline sc

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